

# Overview

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Mitesu

Maximize the introduction of

# Plan for

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January 2025 (Reiwa 7)

Nara Prefecture, Mitesu

## Purpose of Plan Formulation

In recent years, serious natural disasters have increased worldwide due to climate change and extreme weather believed to be caused by global warming. The acceleration of temperature rise shows no signs of stopping, and an era of global boiling has begun, it is said.

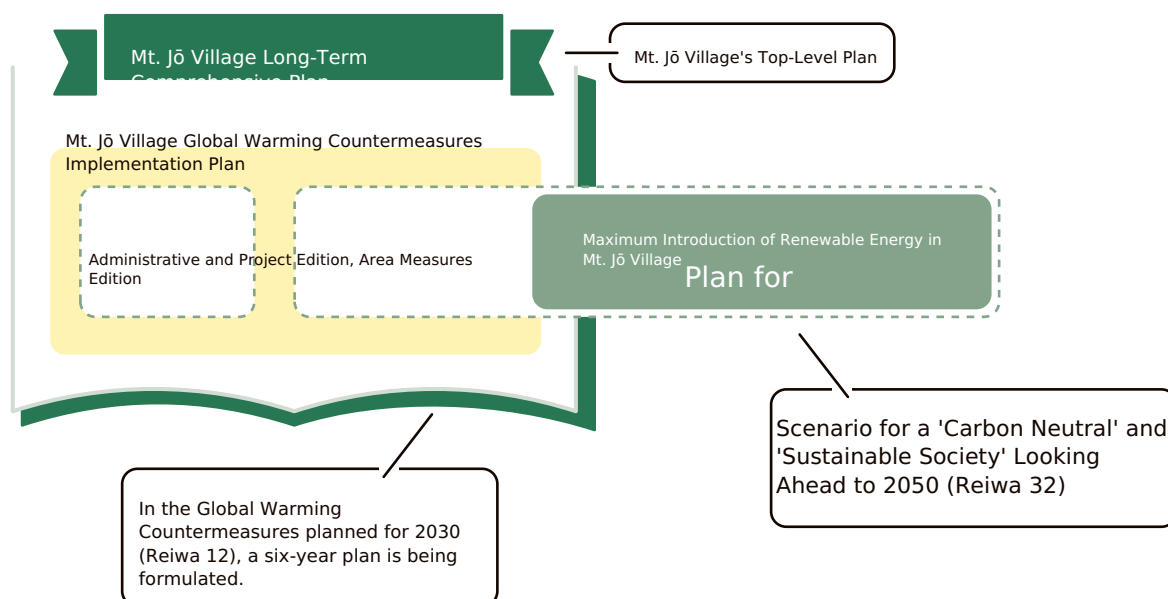
This plan aims to realize a decarbonized society by 2050 (Reiwa 32) by introducing renewable energy tailored to our village's characteristics and undertaking village-wide global warming countermeasures and measures to increase forest absorption, showing a path toward carbon neutrality.

At the same time, we aim for a sustainable villagemanagement of decarbonization that enhances village vitality, well-being of villagers, and secure living, making villagers and future generations happy, unique to Mt. Jō Village (Gojō) and to realize a decarbonized village development.

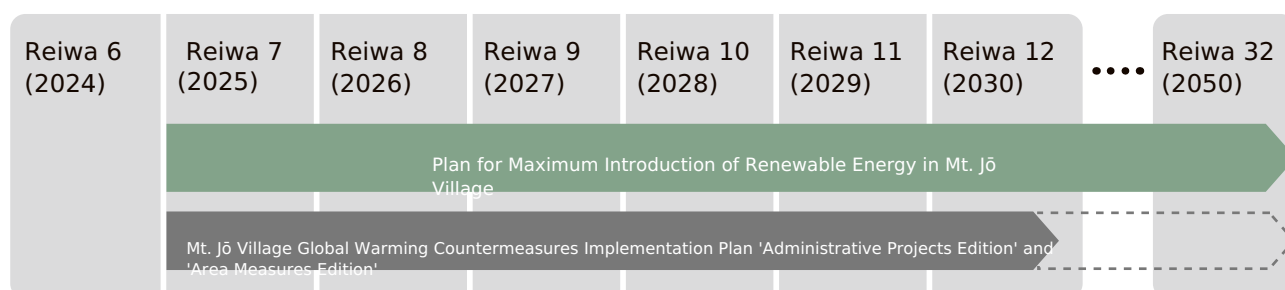
## Position and Period of the Plan

This plan is aligned with national and Nara Prefecture plans, as well as higher-level plans for the village such as the Mt. Jō Village Long-Term Comprehensive Plan and the First Mt. Jō Village Global Warming Countermeasures Implementation Plan, etc. It is also formulated as an integral part of the Global Warming Countermeasures Implementation Plan 'Area Measures' edition.

### Image of Interrelatedness of Environmental Plans



### Plan Period of This Plan



# Basic concepts toward achieving carbon neutrality

## 1) Reduction of energy consumption

Reduce energy consumption through thorough energy conservation, etc.

|          |            |                                                                                                                                                                                                                       |
|----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Examples | Businesses | Review production processes, streamline logistics, introduce energy-efficient equipment, reduce unused electricity, choose low-CO2-emission transportation as much as possible, switch to energy-saving devices, etc. |
|          | Villagers  |                                                                                                                                                                                                                       |

## 2) Energy use transformation

According to how energy is used, change the type of energy used as heat or electricity to achieve greater CO<sub>2</sub> reductions

|          |                              |                                 |
|----------|------------------------------|---------------------------------|
| Examples | Gasoline vehicles (gasoline) | Electric vehicles (electricity) |
|          | Fan heaters (kerosene)       | Wood stove (wood biomass)       |

## 3) Decarbonization of energy

Reduce CO<sub>2</sub> emissions per unit of energy consumption by introducing renewable energy, etc.

|          |                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------|
| Examples | Introduction of solar, wind, hydro, geothermal, and woody biomass that do not emit CO <sub>2</sub> during power generation |
|----------|----------------------------------------------------------------------------------------------------------------------------|

## 4) Sinks and offset measures

While advancing 1-3,

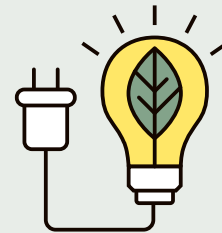
Simultaneously enhance CO<sub>2</sub> absorption through proper forest conservation and management, leveraging our village's forest-rich characteristics for a long-term decarbonization effort

|          |                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------|
| Examples | Proper forest conservation and management,                                                      |
|          | Advancement of absorption measures through maintenance and development of the forestry industry |

### Energy

Villagers, businesses, and government can each take initial actions

Start from there



### Energy creation (new energy)

Popularization of renewable energy that does not depend on fossil fuels



### Carbon offset

The rich forests, a treasure of Mitate Village,

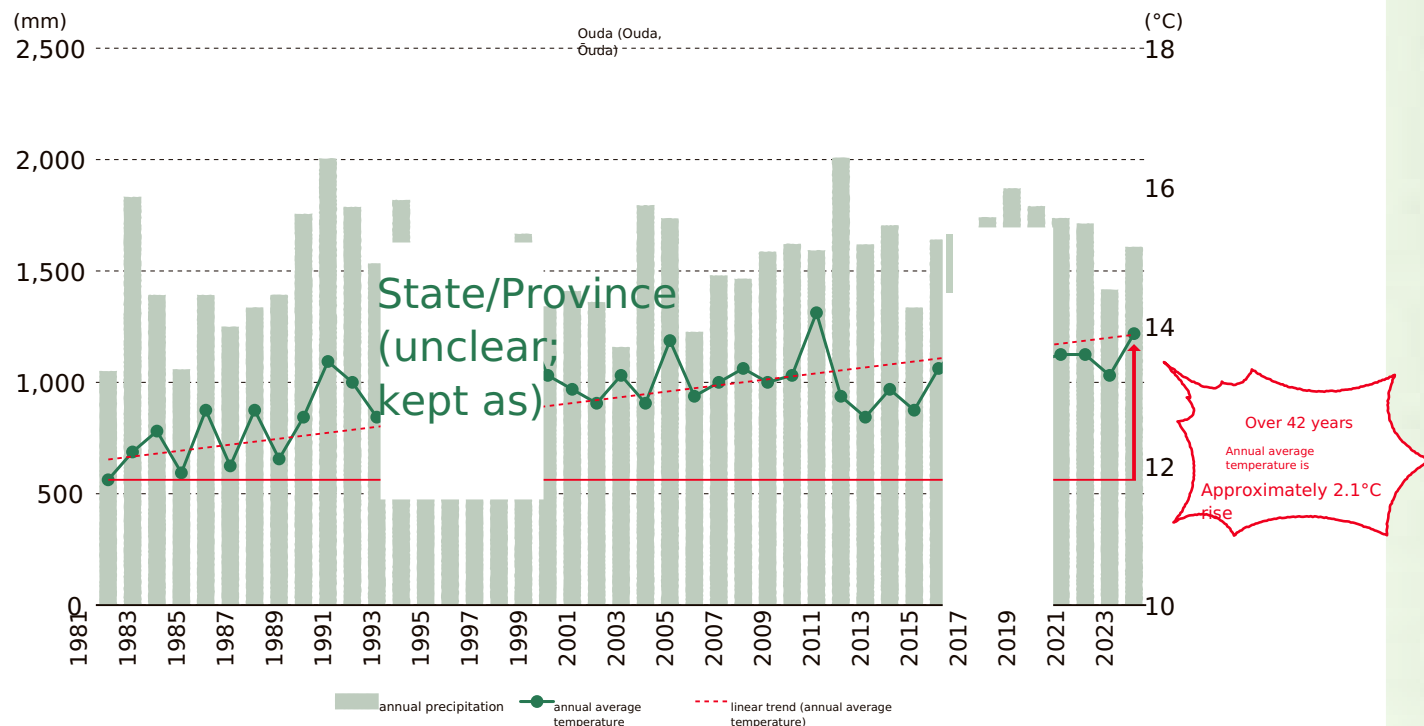
In a long cycle, toward CO<sub>2</sub> sinks



## The situation of global warming surrounding Mitsu Village

### Trends in annual average temperature and precipitation

The following graph shows long-term annual precipitation and annual average temperature from observation sites near Mitsu Village. The average temperature over the years shows an upward trend, indicating warming is progressing.



Source: 'Meteorological Statistics' (Japan Meteorological Agency website); average temperature at the observation site near Mitsu Village 'Ouda'

## Mitsu Village Initiatives

In Mitsu Village, efforts to reduce CO2 emissions include installing solar power for nurseries and housing for young singles. In addition, nurseries have storage batteries installed, and housing for young singles is configured to Nearly Zero Energy House (NearlyZEH) standards. Also, at the tourist facility Mitsue Onsen 'Himeshi no Yu', a wood-fired boiler has been installed to reduce fossil fuel usage.



Mitsu Village-produced firewood materials



Housing for young singles

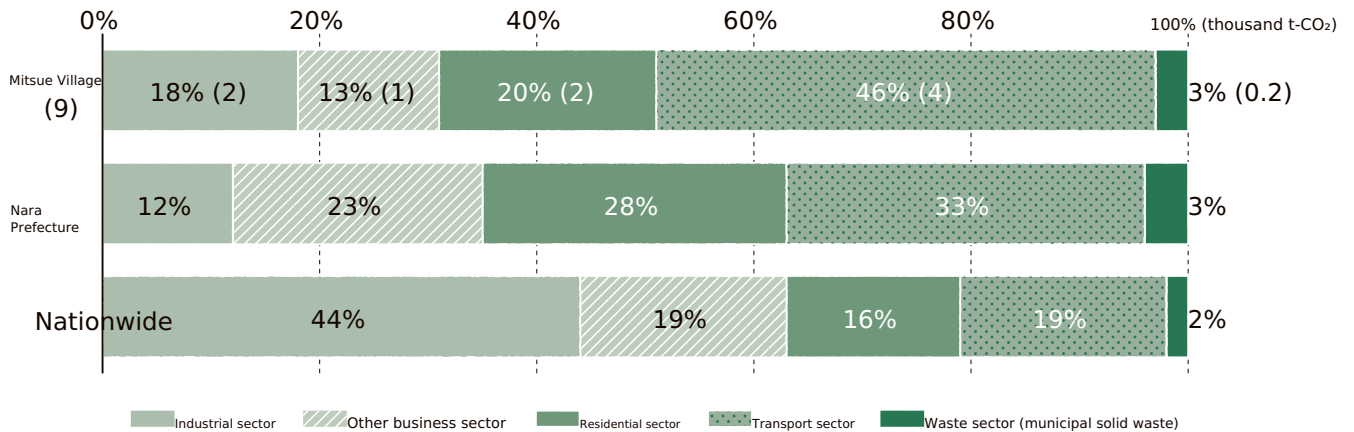


Mitsue Onsen 'Hishi no Yu'

## CO2 emissions in Mikawa Village

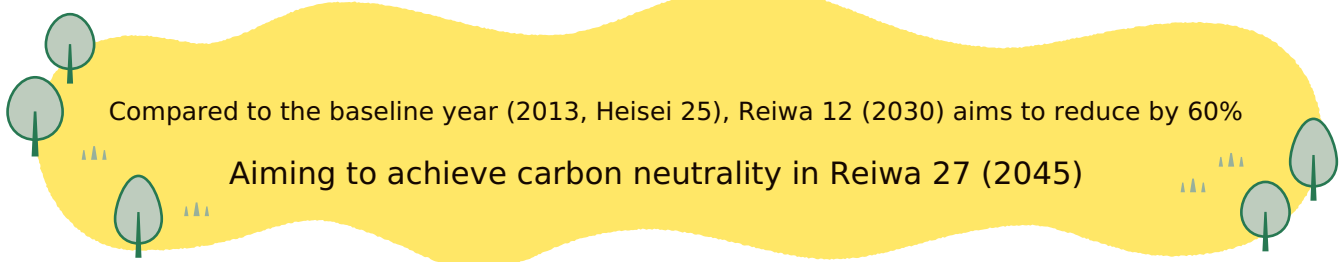
■ Comparison of the composition ratio by department/sector (奈良県 average and national average) (Fiscal year 2021)

Mitsue Village CO<sub>2</sub> emissions amount to 9 thousand t-CO<sub>2</sub>. By sector, the transportation sector is the largest, and in comparison with national and Nara Prefecture averages, the transportation sector is higher and the other business sector is lower.



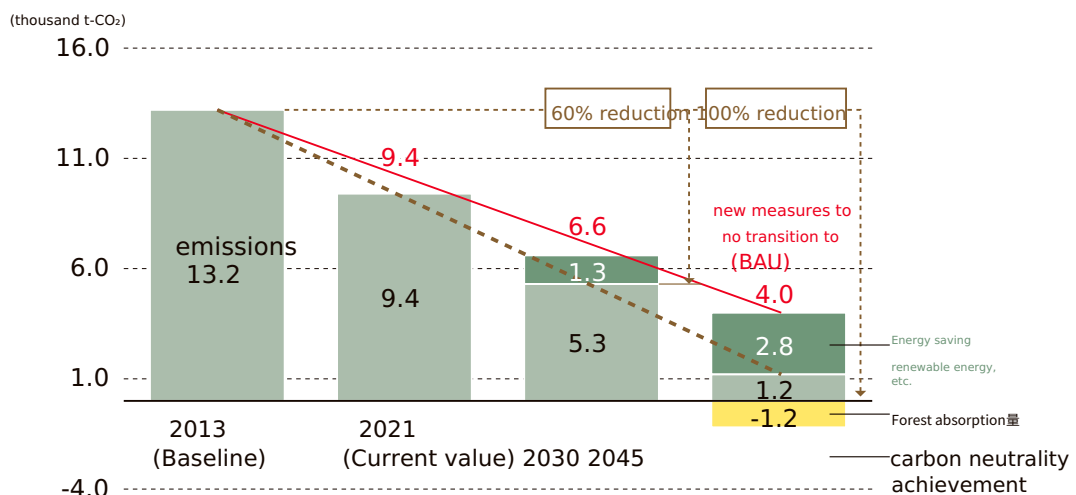
\*Due to rounding, the totals may not add up exactly.

## Mitsue Village CO2 reduction target



In Muesh Village, by promoting the various initiatives described below, we aim to surpass the national target of Reiwa 32 (2050) and pursue carbon neutrality in Reiwa 27 (2045). This includes not only energy saving and energy creation but also the steady development of our village's abundant forests to enhance their function as a CO<sub>2</sub> sink.

■ Mue Village CO<sub>2</sub> Reduction Target Amount



## Three main things we want to tackle (Key Initiatives)

### 1. Local production and consumption of energy unique to Otsuke

目指す方向性

We will focus on creating and effectively utilizing renewable energy that harmonizes with Otsuke Village's geography and natural conditions, so that local production and consumption of energy becomes a value for the village that is utilized and inherited into the future.

- ▶ Otsuke Village's rich nature is a pride and asset to be inherited for the future; therefore, we will produce renewable energy in harmony with nature, with minimal environmental impact.
- ▶ By generating renewable energy within the village, it becomes possible to use it as power during disasters and emergencies. Therefore, we will produce energy primarily at public facilities on a daily basis and build a distributed energy system that can be effectively used in both peacetime and emergencies.

## Key Initiatives

- We will undertake surveys to identify suitable sites and introduce small-scale hydropower equipment by utilizing intake points of simple water supply facilities.
- We will promote the installation of solar photovoltaic facilities on the premises and roofs of public facilities.
- We will build a system to charge batteries and electric vehicles with electricity produced by small hydro and solar power, usable in both peacetime and emergencies.
- We will establish a system that uses V2H (charging/discharging devices) to consume the electricity stored in electric vehicles within buildings.
- By combining these everyday renewable energy production and consumption, we will build a system for effective use of electricity assuming emergencies.





## 2 Utilize the rich forests of Gose Village from a decarbonization perspective

### 目指す方向性

Taking decarbonization as an opportunity, we will strive to further maximize the effective use and enhancement of the village's rich forests, so that this becomes a new added value of the forests, with the village-wide focus on raising awareness of the importance of forest use and promoting sustainable circular utilization.

- ▶ To increase CO<sub>2</sub> absorption from the rich forests that are the property of Gose Village, we will clarify forest ownership and boundaries and advance proper and planned forest maintenance.
- ▶ We will steadily implement the clarifications of forest owners and boundaries and planned maintenance, and if the CO<sub>2</sub> absorption from forests is secured, we will explore introducing a J-Credit system that enables buying and selling as added value of forests.

### Key Initiatives

- To remove barriers to forest maintenance, we will create basic materials such as 'Forest Lot Maps' and promote地籍 surveys, and survey forest owners' intentions to clarify ownership, boundaries, and silvicultural objectives.
- We will support steady forest maintenance and the preparation of forest management plans by forestry management entities.
- We will develop forestry workers equipped with both forest maintenance skills and expertise, and improve the employment environment. We will also focus on the new forestry style of 'self-harvesting forestry' and establish an forestry management style suited to Gose Village's forest environment, while securing and nurturing diverse forestry workers.
- In anticipation of future J-Credit generation and sales, we will work to create a system that consolidates forests into motivated and capable forestry management bodies through collaboration with stakeholders across diverse sectors.



### 3 Increase in value and pride as an environmentally conscious village

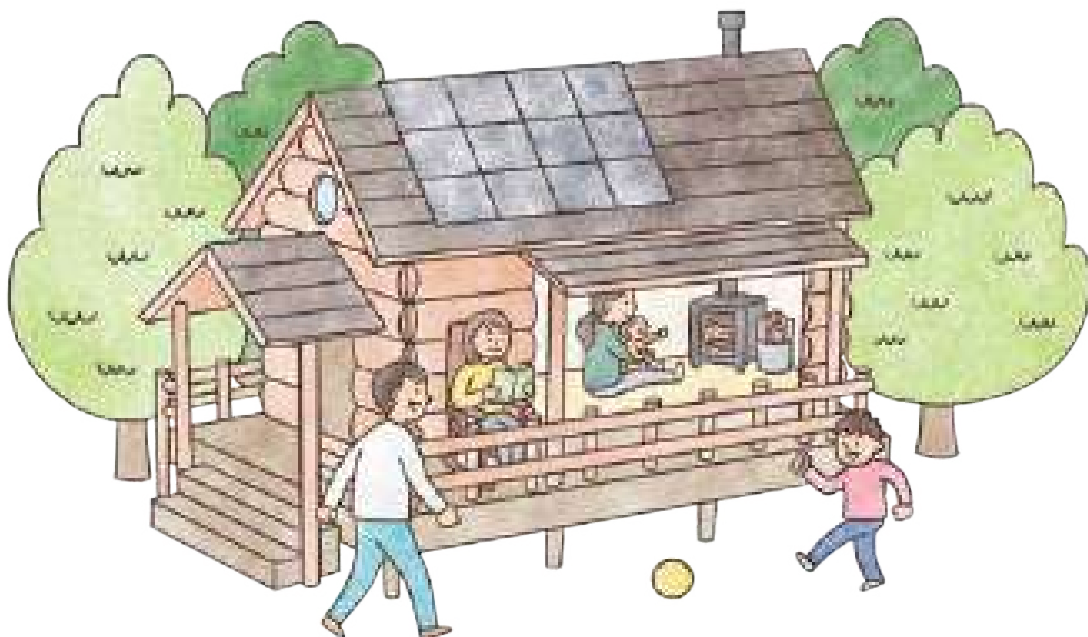
#### 目指す方向性

As the world moves toward carbon neutrality, our village's efforts toward decarbonization will contribute to increasing the village's value, so that all villagers can take pride and act voluntarily, with a focus on branding local resources from a decarbonization perspective and making environmentally considerate initiatives visible.

- ▶ We will brand an environmentally friendly and gentle way of living as a new appeal of Mura, branding it as a decarbonized lifestyle and business style unique to Gojumura, and attract new residents and settlers.
- ▶ Actions by villagers, businesses, and the community that are environmentally considerate will contribute to sustainable village making, and will become a value of Gojurimura that also leads to increased visitors and related population, as we effectively communicate decarbonization initiatives inside and outside the village.

#### Key initiatives

- To attract young people to move and secure successors in the forestry sector, we will brand an environmentally conscious "eco rural living" as a charm of living in Gozumura.
- We will incorporate decarbonization into the use of vacant houses and housing environmental improvement projects, and promote migration and settlement measures that include energy saving and energy creation.
- We will enhance various supports and learning opportunities so that villagers and businesses can actively engage in decarbonization actions and information gathering, and disseminate information through easy-to-understand public outreach media.
- To motivate administration, villagers, businesses, and the community to promote decarbonization together, we will work on making each person's environmentally considerate actions and results visible.





## Collective measures against global warming (regional policy edition)

The forests covering about 90% of the village area and the rich nature that spreads there are the pride of this village and a treasure to be handed down to the next generation. In advancing decarbonization with an eye toward 2050 (Reiwa 32) in this village, it is important to promote energy-saving actions with the community, businesses, and government with ALL Mitake while preserving the natural environment, introduce renewable energy with as little environmental burden as possible, and promote energy self-sufficiency. In addition, by preserving forests and managing them properly, their value as CO<sub>2</sub> absorbers becomes more important.

Promote understanding of decarbonization efforts, behavior change, and system-building so that these initiatives lead to a safe and prosperous life, by implementing comprehensive policies.

### Basic Policy

### Basic Measures

#### 1. Promotion of



Act Okita (note: likely a formatting artifact; translation kept as is)

1. Promotion of Energy-Saving Actions
2. Promotion of High-Efficiency Energy-Saving Equipment and Facilities
3. Improvement of Building Energy-Saving Performance such as ZEB/ZEH

#### 2. Sustainable Village Building through Local Production for Local Consumption of Renewable Energy



1. Promotion of Renewable Energy Introduction
2. Formation of an Independent and Distributed Energy Society for Disaster Preparedness

#### 3. Promotion of Low-Carbon

Ushiku 7 8Koku Kai (note: likely formatting artifact; translation kept as is)

1. Promotion of Electric Vehicles (EV, PHV, PHEV, FCV)
2. Promotion of Low-Carbon Mobility

#### 4. Formation of a



Pass/Teds (note: likely fragment)

1. Reduction of CO<sub>2</sub> Emissions by Reducing Waste
2. Addressing Solar Panel Disposal Issues

#### 5 Decarbonization Efforts by All Actors



1. Transition to Decarbonized Lifestyles and Business Styles
2. Promotion of Global Warming Prevention Measures

3. CO<sub>2</sub> Absorption by Forests

# Shifty ① Okadan

Basic Policy

## 1

### Promotion of energy-saving measures

To promote energy efficiency, it is important to shift to a low-carbon lifestyle that balances a healthy and comfortable life by not just enduring, but by obtaining diverse information and taking action.

#### 1. Promotion of energy-saving actions



Government measures

- We will promote the spread of energy-saving actions in daily life.
- We will raise awareness for the spread of energy-saving actions recommended by the government, such as 'Deco-katsu'. In addition, Misuke Village will register in support of the 'Deco-katsu Declaration'.

- To reduce CO<sub>2</sub> emissions, promote the spread of delivery boxes and leave-at-home delivery.
- Considering energy demand pressures, we will promote Demand Response, a system that issues timely electric-saving requests to power users according to demand, including exploring the use of disaster avoidance lines and wired broadcasts.



Villagers and businesses,  
Community efforts

- We will keep energy-saving actions in mind.
- We will understand the national movement 'Deco-katsu' promoted by the government and start with what we can do.
- In places where people gather, such as communities and workplaces, we will hold study sessions on energy-saving that balance a healthy and comfortable life.

#### 2. Promotion of high-efficiency energy-saving equipment and facilities



Administrative efforts

- We will disseminate information to villagers and to businesses about various support systems for introducing energy-saving devices and efficient operation methods.
- When purchasing new products, we will promote the purchase of high-efficiency appliances with high energy-saving effects, including raising awareness of energy-saving labels.



Villagers and businesses,  
Community efforts

- We will promote LED lighting.
- When replacing air conditioners, refrigerators, TVs, etc., we will choose high-efficiency devices with reference to energy-saving labels.

### 3. Improvement of energy-saving performance of buildings such as ZEB and ZEH



Government initiatives

- When constructing new buildings such as houses and offices or proceeding with renovations, we will promote the spread of ZEB (Net Zero Energy Building) and ZEH (Net Zero Energy House). In addition, we will provide information to use building energy efficiently by utilizing systems such as HEMS (Home Energy Management System).
- During extensions or renovations of houses and offices, we will promote high insulation of roofs, exterior walls, and fittings, and encourage economically efficient insulation of individual rooms to popularize proactive energy-saving standards, including providing various subsidy information.
- Homes with high insulation not only stay cool in summer and keep heat from escaping in winter, but also help prevent condensation and mold, reduce winter heat shocks, stabilize blood pressure, and contribute to overall health; we will clearly inform about such information.
- We will raise awareness of LCCM housing (Life Cycle Carbon Minus Housing) concepts and promote efforts to reduce CO<sub>2</sub> emissions during construction and disposal.



Villagers and  
businesses,  
Community initiatives

- When constructing new buildings or making renovations, we will obtain information on ZEB, ZEH, and HEMS to ensure high energy-saving performance.
- Even if not undergoing large-scale renovations, we will address insulation measures for gaps in houses because insulation effects are high.





## Basic Policy

# 2

## Sustainable village development through local production for local consumption of renewable energy

In addition to traditional large-scale, centralized energy, we will utilize locally produced renewable energy as 'distributed energy' to diversify energy supply risks and reduce CO<sub>2</sub> emissions. Also, considering economic circulation, regional revitalization, and other aspects of achieving a sustainable society, we will promote the introduction of renewable energy tailored to the village's characteristics.

### 1. Promotion of Renewable Energy Introduction



#### Government Initiatives

- By providing information and promoting subsidies related to solar power generation and storage batteries, we will promote the installation of solar panels and storage batteries on buildings by businesses and villagers.
- We will raise awareness of PPA projects that do not require upfront costs.

- We will collect information on the introduction of renewable energy using new technologies, including perovskite.
- We will raise awareness of solar carports and other self-consumption solar power generation equipment, not only on buildings.
- We will identify suitable locations and promote the introduction of renewable energy, including small hydropower, to maximize renewable energy potential according to regional characteristics.



#### Villagers and businesses, Local Initiatives

- We will promote the use of self-consumption renewable energy, such as installing solar on residential roofs. We will also utilize PPA projects that have no upfront costs.
- We will consider various forms of renewable energy usage, such as installing solar carports in parking lots.
- We will participate in opportunities to gain knowledge on effectively utilizing renewable energy.



## 2. Preparedness for Disasters through the Formation of an Autonomous and Distributed Energy Society



Governmental  
Initiatives

- Treat electric vehicles (EVs) and plug-in hybrid vehicles (PHEVs) as “mobile storage batteries,” and consider a system that charges these EVs with surplus daytime power from locally generated renewable energy so that, in an emergency, the entire region can effectively utilize energy through EVs.
- Combine storage equipment such as EVs, PHEVs, and storage batteries with power generation facilities like solar power to build a resilient system for disaster scenarios.
- To develop disaster-resilient village structures that can supply power to major facilities such as public buildings and clinics, normally connected to the transmission and distribution network, but in emergencies disconnect the area from the network (off-grid technology) to enable energy self-sufficiency through distributed generation.



Villagers and  
businesses,  
Regional Initiatives

- Effectively introduce electric vehicles and V2H (charging/discharging devices) to use the stored electricity in EVs for households, thereby utilizing energy efficiently.
- To effectively use power generated from solar photovoltaic systems, consider introducing storage batteries and promote local production for local consumption of energy.



## 3

## Promotion of low-carbon mobility

This village has a high share of CO2 emissions from the transportation sector, accounting for more than one-third of total emissions. We will promote early adoption of electric vehicles, raise awareness of eco-driving, and enhance public transportation and on-demand transport, while devising low-carbon mobility options that also improve daily life convenience.

### 1. Promotion of Electric Vehicles (EV, PHV, PHEV, FCV)



Administrative initiatives

- Ahead of the planned abolition of new gasoline car sales in Reiwa 17 (2035), we will promote the spread of electric vehicles (EVs, PHVs, PHEVs, FCVs).

- Public facilities such as the town hall and roadside stations will be equipped with EV chargers as a priority.

- By charging with electricity generated from renewable energy, zero CO2 emissions during driving will be achieved, promoting zero-carbon driving.
- For vehicles where electrification is deemed difficult, we will promote switching to clean fuels such as biofuels and synthetic fuels (e-fuels).



Villagers and businesses, Regional initiatives

- When purchasing a vehicle, consider introducing electric vehicles (EVs), plug-in hybrid vehicles (PHEV), fuel cell vehicles (FCV), etc.

### 2. Promotion of low-carbon mobility



Administrative initiatives

- By improving public transportation and on-demand mobility, we will promote daily travel options with minimal CO2 emissions for commuting, schooling, and other routine movement.
- We will promote eco-driving to encourage fuel-efficient driving.



Villagers and businesses, Regional initiatives

- When using a car, practice fuel-efficient driving (eco-driving).
- Choose low-CO2-emission mobility options such as on-demand transport and public transit.



# 4

## Formation of a circular society

Reducing waste leads to a reduction in CO<sub>2</sub> emissions from incineration. We will promote reduction of plastic waste, which has high CO<sub>2</sub> emissions, and actions to prevent food loss. At the same time, we will work on reuse of waste and effective use of resources. Through these efforts, we will promote the formation of a circular society.

### 1. Suppression of CO<sub>2</sub> emissions by reducing waste



#### Administrative measures

- To suppress waste generation and incineration, we will promote resource recycling of plastics and other materials.
- We will promote efforts to reduce food loss through takeout of leftovers (mottECO).
- To reduce food loss, we will promote the spread of food drive activities and support for food bank activities.
- We will promote composting of raw household garbage and efforts toward resource recovery for waste generators, and raise awareness.
- We will promote the 4R activities (Refuse, Redeye, Reuse, Recycle) that lead to waste prevention and reduction/resource recovery.
- We will distribute pamphlets and hold outreach lectures to provide information and raise awareness on waste reduction and recycling.
- We will promote using reusable bags and green purchasing.
- To reduce paper waste such as old paper, we will promote support for community collection and awareness of paper recycling.
- We will promote efforts by businesses to reduce waste and recycle resources.



#### Residents, businesses, Community initiatives

- We will strive to prevent and reduce waste generation.
- We will thoroughly separate household and business waste to promote resource recovery.
- We will strive for reuse by passing on things that are still usable to others.
- We will actively choose recycled products.
- We will bring and use reusable bags to reduce plastic shopping bags and over-packaging.
- To reduce plastic, we will implement light packaging and reduce disposable containers.

### 2. Addressing disposal issues of solar panels



#### Administrative measures

- Regarding solar panels, based on the government's guidelines for promoting recycling of solar power facilities, we will disseminate information to ensure proper disposal.



Basic policy

# 5

## Efforts for decarbonization by all主体

To realize a carbon-neutral village that suits Mitsue-mura with residents, businesses, and administration working together, we will promote initiatives that lead to changes in behavior and awareness of residents and businesses, pursue community development from a long-term perspective, utilize energy from renewable sources, and implement diverse global warming prevention measures. We will also work on forest management to enhance forest absorption capacity.

### 1. Transition to Decarbonized Lifestyles and Business Styles



#### Government Initiatives

- To promote the transition to decarbonized lifestyles, we will advance environmental learning in collaboration with educational institutions and lifelong learning efforts, and promote Education for Sustainable Development (ESD).
- To promote local economic circulation and a thriving local economy, we will explore creating new products and brands through decarbonization and provide opportunities to brainstorm ideas for sustainable village-making that leverages decarbonization.
- We will promote effective promotion so that decarbonization efforts enhance Mitsu Village's image and foster pride and attachment (civic pride) among residents.
- Towards the transition to decarbonized lifestyles, we will raise awareness of information disseminated by the national and prefectural governments and various apps that lead to decarbonization, provide information by leveraging opportunities where people gather in the village, and promote behavioral changes toward decarbonization through various means.
- For decarbonization in the agricultural sector, we will consider and support efforts to expand organic cultivation areas and reduce the use of chemical fertilizers, and promote the smart use of farming equipment.



#### Residents, businesses,

#### Local initiatives

- In daily life and business, we will practice actions we can take ourselves, such as choosing environmentally conscious actions and products/services to address global warming.
- Actively participate in environments and events where you can learn about global warming countermeasures.
- We will strive to reduce chemical fertilizers and improve agricultural management efficiency through the smart use of farming equipment.

## 2. Promotion of measures to prevent global warming



Administration's  
Initiatives

- In bidding for public works and procurement and service provision ordered by the village, we will study systems to evaluate businesses undertaking decarbonization, such as procurement under the Green Procurement Act and the Environmental Consideration Contract Act.

- We will raise awareness among residents and businesses to promote procurement of low-carbon electricity derived from renewable energy.
- We will investigate and study measures such as building VPP (Virtual Power Plant) using IoT technologies (including blockchain) and storage systems in collaboration with private businesses to stabilize the power grid and level demand, and consider expanding renewable energy adoption and further energy conservation.



Residents and  
businesses  
Community initiatives

- We will strive to choose electricity sourced from renewable energy.

## 3. CO2 absorption by forests



Administration's  
Initiatives

- Forest absorption contributes to reducing actual CO2 emissions and has considerable potential, including carbon credit trading. We will promote planned forest management such as thinning to secure forest absorption amounts and consider introducing and utilizing J-Credits.

- To restore and protect healthy forests, we will promote forest management using the national Forest Environment Transfer Tax and Nara Prefecture Forest Environment Tax projects.
- We will study the use of carbon offsets through inter-municipal cooperation, offsetting the carbon absorption of your village's forests via the J-Credit system with municipalities in metropolitan areas where achieving carbon neutrality is unlikely.
- To promote local production for local consumption of energy, we will consider using thinning wood from village forest management as fuel and promote the spread of wood stoves.



Residents and  
businesses  
Community initiatives

- We will consider installing wood stoves.
- We will cooperate to clarify forest ownership and boundaries and conduct surveys of intentions to enable planned forest management.
- We will work to discover human resources involved in forest-related businesses.

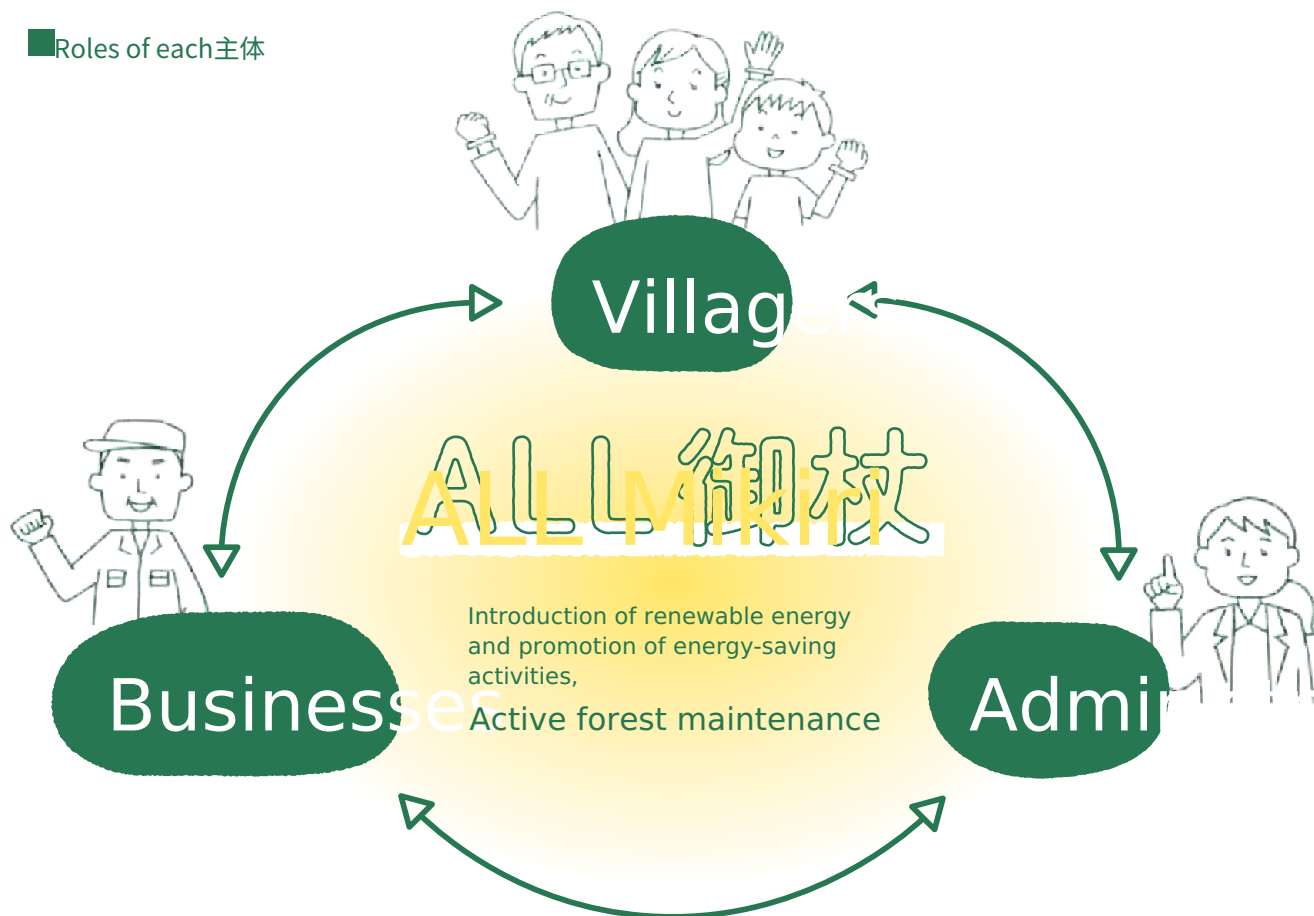
## List of related indicators

| Survey indicators                                                                                                      | [Target values (implementation rate)] |                     |                                 |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------|---------------------------------|
|                                                                                                                        | Current<br>Year 2024                  | Target<br>Year 2030 | Target<br>Year 2045             |
| Proportion of summer air conditioning set to 28°C or higher                                                            | 19.9%                                 | 41.6%               | 100%                            |
| Proportion of cleaning air conditioner filters 1-2 times per month                                                     | 14.2%                                 | 38.6%               | 100%                            |
| Proportion of kerosene fan heaters set to 20°C or below                                                                | 36.0%                                 | 66.9%               | 100%                            |
| Proportion of not leaving hot water running when washing dishes by hand                                                | 60.4%                                 | 88.0%               | 100%                            |
| Proportion of not overpacking the refrigerator                                                                         | 51.7%                                 | 84.4%               | 100%                            |
| When using a gas stove, ensure the flame does not extend beyond the pot bottom                                         | 53.2%                                 | 69.3%               | 100%                            |
| Proportion of adjusting                                                                                                |                                       |                     |                                 |
| Proportion of implementing LED lamp bulbs (bulb-shaped)                                                                | 72.8%                                 | 79.2%               | 100%                            |
| Proportion of high-efficiency water heaters (EcoCute, EcoJozu, etc.)                                                   | 33.2%                                 | 39.1%               | 55.0%                           |
| Proportion of introduction                                                                                             |                                       |                     |                                 |
| Energy-saving housing (houses with high insulation and heat-shielding performance in walls and windows)                | 13.8%                                 | 22.9%               | 45.0%                           |
| Proportion of newly built                                                                                              |                                       |                     |                                 |
| Renovation to energy-saving housing (with high insulation/solar heat mitigation performance)                           | 7.8%                                  | 13.5%               | 30.0%                           |
| The proportion of doing                                                                                                |                                       |                     |                                 |
| The percentage of introducing HEMS (Home Energy Management System)                                                     | 0.5%                                  | 6.6%                | 25.0%                           |
| Basis: The percentage of using solar power at home                                                                     | 4.1%                                  | 6.0%                | 14.0%                           |
| Hon                                                                                                                    |                                       |                     |                                 |
| Hou                                                                                                                    |                                       |                     |                                 |
| Shin                                                                                                                   |                                       |                     |                                 |
| Utilizing renewable energy and storage batteries, electric vehicles, etc.                                              |                                       |                     | - - One site within the village |
| 2 Building resilient systems                                                                                           |                                       |                     |                                 |
| The percentage of introducing hybrid cars, electric vehicles, etc.                                                     | 30.4%                                 | 58.2%               | 100%                            |
| Basis                                                                                                                  |                                       |                     |                                 |
| Hon                                                                                                                    |                                       |                     |                                 |
| Policy: Gradually accelerating the car, etc.                                                                           |                                       |                     |                                 |
| Shin                                                                                                                   |                                       |                     |                                 |
| 3 The percentage of aiming for fuel-efficient driving                                                                  | 64.1%                                 | 91.7%               | 100%                            |
| Estimated total for passenger and freight gasoline vehicles                                                            |                                       |                     |                                 |
| The percentage of carrying a reusable bag and not buying plastic bags                                                  | 71.0%                                 | 80.0%               | 100%                            |
| Basis                                                                                                                  |                                       |                     |                                 |
| Hon                                                                                                                    |                                       |                     |                                 |
| Hou                                                                                                                    |                                       |                     |                                 |
| Shin                                                                                                                   |                                       |                     |                                 |
| 4 Gokujou Village waste emissions (FY2022 results)                                                                     | 340 t                                 | -                   | 170 t                           |
| Proportion of chemical fertilizer use in agriculture                                                                   | 90.0%                                 | -                   | 45.0%                           |
| Basis                                                                                                                  |                                       |                     |                                 |
| Hon                                                                                                                    |                                       |                     |                                 |
| Method                                                                                                                 |                                       |                     |                                 |
| Pointer                                                                                                                |                                       |                     |                                 |
| 5 The percentage of people who are contracted with an electric utility company and a renewable energy-based power menu | 8.8%                                  | 30.0%               | 100%                            |
| Forest absorption by forest maintenance                                                                                | 0.3 thousand                          | 1.8 thousand tons   | 2.7 thousand tons               |

## ALL Mikiri (Initiatives) (Promotion System)

To achieve the CO<sub>2</sub> emission reduction targets outlined in this plan, all主体s will cooperate with each other and the community will unite to promote 'ALL Mikiri'.

### Roles of each主体



### Promotion Structure



- A council consisting of experts, regional representatives, and business operators will be held regularly to review, evaluate, and propose revisions to the plan. In addition, opportunities for consultation and exchange of opinions will be provided for various measures and initiatives with diverse stakeholders, not limited to the council, to enhance the plan's effectiveness.

- Promotion Headquarters: Led by the village chief as head and the deputy village chief as deputy head, with other members such as managers to organize the promotion headquarters, formulating and reviewing the plan and conducting comprehensive promotion inspections.
- Promotion Officers: Assign at least one 'Promotion Officer' in each department. The 'Promotion Officer' will oversee the plan's promotion and progress, coordinating with the secretariat to advance the plan comprehensively.
- Secretariat: Based in the Resident Life Division, responsible for promoting the entire plan and monitoring progress, and managing overall progress.



# Glossary

## Eco Driving

Driving techniques and practices that reduce fuel consumption and CO2 emissions, such as gradual acceleration and braking and maintaining ample following distance, to help prevent global warming.

## Energy-Intensive Unit

The total amount of energy consumption, such as electricity and heat (fuel) required to produce a fixed quantity of products, representing energy efficiency.

## Greenhouse Gases

Gases that cause greenhouse effect dispersed in the atmosphere. In addition to CO2 and CH4 (methane), fluorinated gases and others show increasing atmospheric concentrations due to human activities.

## On-demand Transportation

Vehicles that operate by setting the optimal route using AI and other technologies according to user reservations.

## Carbon Offsetting

The concept of compensating for unavoidable greenhouse gas emissions (such as CO2) by investing in activities that reduce or offset those emissions.

## Carbon Neutral

To achieve net-zero greenhouse gas emissions, including CO2; reduce emissions and offset absorbed amounts by forests and other sinks.

## Renewable Energy (RE)

Defined as energy sources that can be used sustainably: solar, wind, hydro, geothermal, solar heat, atmospheric heat, and biomass. Renewable energy resources do not deplete and can be used repeatedly; they emit little to no CO2 during generation or use.

## Food Loss

Food that could be eaten but is discarded, such as leftovers, items bought in bulk on sale that could not be eaten, or vegetables with too thick skin.

## J-Credits

A system whereby greenhouse gas reductions/absorption achieved through renewable energy adoption, energy-efficient equipment, or forest management (afforestation, thinning) are quantified, certified by the government as tradable 'credits'.

## Self-sufficient/Distributed Energy Society

A society where small distributed generation facilities meet each consumer's electricity needs, efficiently integrated with the grid to ensure stable power supply even during outages when the grid is unavailable.

## Creative Energy (Energy Creation)

Producing renewable energy.

## Electric Vehicle (EV)

Abbreviation for Electric Vehicle (EV); unlike gasoline cars that burn fuel in an engine to drive the vehicle, EVs emit no exhaust gases and have significantly reduced running noise.

## V2H (Charge-Discharge Device)

A system that allows the electricity stored in the batteries of EVs or PHEVs to be used at home or facilities.

## Wood-based Biomass

Biomass resource made from wood; includes forest residuals such as branches and leaves, bark and sawdust from sawmills, demolition materials from houses, and pruned branches of street trees.

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